

Aditya Kulkarni

adityakulkarnius@gmail.com | (281) 876-7066 | linkedin.com/in/aditya-kulkarni-355b81217 | github.com/Aditya-k24 | kulkarniaditya.com | Raleigh, NC

SUMMARY

Applied AI engineer specializing in LLM evaluation, agentic orchestration, and RAG architectures. Built a vendor-neutral LLM eval harness tracking hallucination rate, factual consistency, p50/p95 latency, and cost-per-call across Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro with paired bootstrap significance testing -- direct hillclimbing experience. Shipped multi-model LLM features via MCP tool handlers and Temporal.io agentic orchestration at Isomer AI (394 commits, 7 deployments). MS CS at NC State (May 2026). Startup pace: rapid prototyping, direct user signal feedback loops, bias for shipping.

EXPERIENCE

- Product Engineer Intern

Isomer AI

May 2025 - Present

 - Architected multi-model LLM routing across Claude, GPT-4o, and Gemini (8+ models) via MCP tool handlers; built prompt architectures, system instruction sets, and Temporal.io durable agentic workflow orchestration with fault-tolerant retries and saga-pattern rollback across 7 production deployments (394 commits).
 - Shipped 10+ backend features in TypeScript, Node.js, Express, PostgreSQL, Redis, and Docker; built document processing pipelines (PDF generation, email parsing via Microsoft Graph API, S3 storage), RBAC, Auth0 authentication, real-time event pipelines (Pusher), and CI/CD via GitHub Actions with Sentry observability.
- Graduate Research Assistant

NC State University

Apr 2025 - Present

 - Designed and deployed a cloud-native genomic data platform on Google Cloud Run serving 500GB of datasets to 100+ researchers; improved ML pipeline accuracy by 12% via hyperparameter tuning and time-series analysis.
- Software Developer Intern

Mphasis (HP Company)

Jun 2024 - Aug 2024

 - Built AI-driven workflows including a health claim processing tool using OCR, NLP, and ML classification models improving accuracy by 25%; automated CI/CD on AWS and GCP, integrated Grafana monitoring, and supported model validation and red-teaming.
- AI/ML Intern

Capgemini

Jun 2023 - Aug 2023

 - Built distributed data pipelines using Hadoop and Apache Spark for large-scale batch processing and Apache Kafka streaming for real-time ingestion, reducing processing time by 15% and end-to-end latency by 20%.

PROJECTS

- LLM Eval Harness

Evaluation / Hillclimbing

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro on 2,000 frozen examples across grounded QA, multi-hop QA, and claim verification. Tracked hallucination rate, factual consistency, abstention accuracy, p50/p95 latency, token throughput, and cost-per-call. Paired bootstrap CIs, McNemar test, and Holm correction for statistically valid model selection. DuckDB + Parquet; Streamlit regression dashboard.

Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker
- AlgoPulse -- Agentic Orchestration

Agentic / Orchestration

Re-architected from API polling to event-driven streaming after diagnosing throughput bottleneck under load; Temporal.io durable workflow state with fault-tolerant retries, activity checkpointing, and saga-pattern rollback; Apache Kafka for partitioned worker coordination. Reached 2,000 async LLM inference req/sec with zero message loss.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka
- Glance AI -- Compliance RAG System

RAG / Grounding

Privacy-first RAG system for fintech and healthcare with per-tenant data isolation at the vector retrieval layer; prevents cross-tenant context leakage and LLM hallucinations in regulated environments. Deterministic output architecture with grounded citations enforces responsible AI principles for compliance-sensitive financial workflows.

Python, LLMs, Vector Databases, RAG, Privacy-First Architecture
- VectorLift -- Semantic Search & Retrieval

NLP / Ranking

Neural IR pipeline on MS MARCO: BM25 sparse baseline, dense bi-encoder retrieval (sentence-transformers, in-batch negatives), cross-encoder re-ranker; Kafka-driven indexing into Elasticsearch + Qdrant; FastAPI serving with per-stage latency tracing; NDCG@10/MRR@10 evaluation across 6 pipeline configurations.

Python, PyTorch, Elasticsearch, Qdrant, Kafka, FastAPI, Prometheus, Grafana, Docker

EDUCATION

- North Carolina State University -- MS Computer Science

Aug 2024 - May 2026

Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms
- Mumbai University -- BTech Computer Science | CGPA: 9.32/10

Aug 2020 - May 2024

Machine Learning, Big Data Infrastructure, DS & Algorithms, Blockchain

SKILLS & AWARDS

AI/ML: LLMs, MCP, RAG, LLM Evaluation, Prompt Engineering, LangChain, PyTorch, Hugging Face, NLP, OpenAI/Anthropic/Google SDKs | **Languages:** Python, TypeScript, JavaScript, Java, SQL, C | **Backend / Infra:** Node.js, FastAPI, Express, Docker, Kafka, Temporal.io, AWS, GCP, CI/CD | **Data:** DuckDB, Parquet, PostgreSQL, MongoDB, Redis, Elasticsearch, Qdrant, Prometheus

HackNC24 Top 10 (100+ teams, 48h) | **Sanjeevani Top 1%** (12 of 1,100+) | **IEEE Published** -- Mapify (500+ attendees) | **Google DSC Lead** (500+ students)